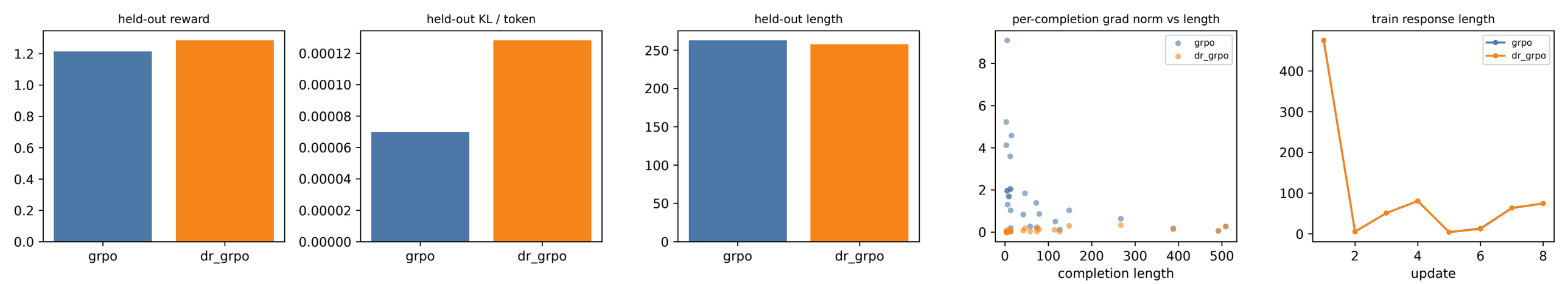
held-out reward
1.2
1.0
0.8
0.6
0.4
0.2
0.0
grpo
dr_grpo
held-out KL / token
0.00012
0.00010
0.00008
0.00006
0.00004
0.00002
0.00000
grpo
dr_grpo
held-out length
250
200
150
100
50
0
grpo
dr_grpo
per-completion grad norm vs length
grpo
dr_grpo
8
6
4
2
0
0
100
200
300
400
500
completion length
train response length
grpo
dr_grpo
400
300
200
100
0
2
4
6
8
update